

Implicit instruction in the OR: Mapping out strategies of implicatures

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Slides:



Echo (1900) by Talbot Hughes

Summary

- ▶ Surgeons teaching residents in the OR
- ▶ Implicit instruction is very common. Why?
- ▶ Which strategies of implicatures do surgeons use?
- ▶ Three main strategies
 - ▶ Gricean relevance
 - ▶ Social connection
 - ▶ Modal embedding
- ▶ Each corresponds to a different pedagogical value

Implicit instruction is very common

(1) explicit instruction

Nick the fascia superior

Case 6 - 05:32

→ [nick the fascia superior]

(2) implicit instruction

So, you wanna go as lateral as you can

C8 - 19:43

→ [go as lateral as you can]

Implicit instruction is very common

High percentage
of instruction is
implicit

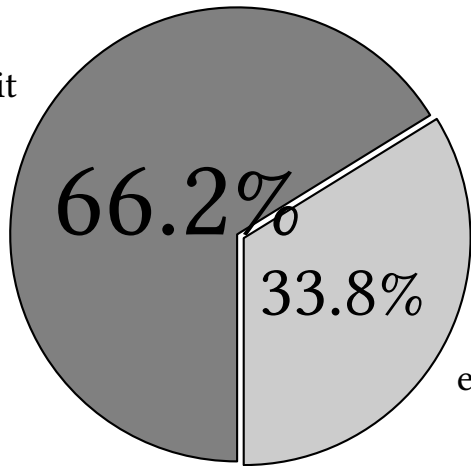
Mahadev et al. 2023

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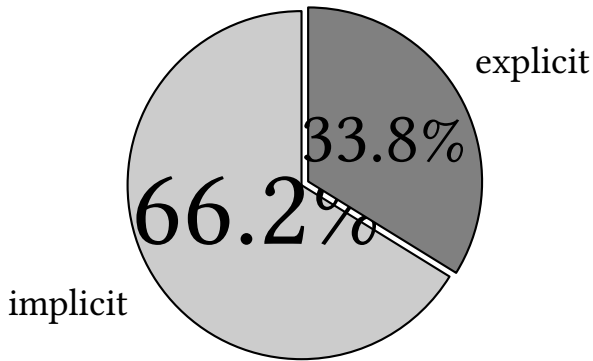
implicit



explicit

Research questions

- ▶ Clearly it's a common technique, but why?
- ▶ First step to that:
What kinds of implicatures do attendings make?
- ▶ Classify implicatures



Methods

- ▶ Six video-recorded surgical teaching cases
319 minutes in all
- ▶ Transcripts made by hand
- ▶ Instruction coded as direct (D) or indirect (I)
- ▶ Indirect (n=656): Implicature types
- ▶ Count up types

6	2612	observation	Q	I	Pretty sure?					
6	2739	steps		I	The question is when we get it what are we gonna do.... Because this is gonna be stiff....					
6	2749	obligation		I	You need to get that way and the rest will be....					
6	2803			D	hold down					
6	2806	steps		I	...the cervix is gonna be....					
6	3021	attitude softening		I	I think that's what you're going to have to do					
6	3021	obligation		I	I think that's what you're going to have to do			two for one		
6	3048	ability	Q	I	Can you somehow...are we gonna be able to do this...cut here and here...					
6	3133	ability	Q	I	can you bring that up, get that up					
6	3200	ability		I	If we can get this back....					
6	3200	conditional		I	If we can get this back....			two for one		
6	3231			D	And then put the lane on and get the ligature just for the back lane just one side at a time					
6	3231	vague assent		I	okay yep					
6	3305	obligation		I	G You have to clamp both sides					
6	3307	conditional		I	G if you just clamp this side and you tie it you're gonna bleed back					
6	3307			D	And then cut...and then cut and tie...					
6	3504	ability	Q	I	Can you elevate that fibroid?					
6	3553	obligation		I	You need to see your pedicle					
6	3553	obligation		I	You just have to hug this fibroid with your bite[?].					
6	3634	what you want		I	There you got that's what you wanna do right there					

Questions (Q) and Generic (G) instructions were also marked, though not counted up for this study

Three classes of implicature

- ▶ **Gricean relevance** Grice 1975
Partial instruction that lets listener infer full instruction
- ▶ **Social connection**
- ▶ **Modal embedding**

Vague assent 21.3%

Surgeon uses words that could mean “stop” or “keep going”

- a) C6 06:50 — **Okay.**
- b) C2 07:23 — Now start cutting, my friend. **Yes. Yes. Oh, yes. Perfect.**
- c) C8 19:20 — **Perfect.**

Steps 11.5%

Surgeon explains the next step
often with *now, then, going to*

- a) C2 09:32 — You're gonna use your PK, essentially like a rake.
- b) C1 14:45 — And **then** we cut towards the umbilicus.
- c) C8 56:30 — And **then** your hand kind of goes back.

Ellipsis 6.0%

Surgeon uses nouns or adverbs without saying what to do.

- a) C8 43:50 — **Suction.**
- b) C3 41:41 — **Kelly, on the right, here.**
- c) C8 17:50 — **Granny knots, please. Square knots stink.**

Observation 7.5%

Surgeon makes pertinent statement to elicit an action or (often) avoiding an action.

- a) C8 14:13 — And where people make a mistake is going right into the areolar tissue right here.
- b) C3 14:01 — No, the blade might come out.
- c) C2 17:32 — That's counter-clockwise.

Three classes of implicature

- ▶ **Gricean relevance**

Partial instruction that lets listener infer full instruction

- ▶ **Social connection**

Instruction is stated but embedded under 1st/2nd person pronouns

- ▶ **Modal embedding**

What you want 9.3%

Surgeon speaks for resident's mind about doing or not doing something.

- a) C8 25:20 — **What you wanna do is dissect along here.**
- b) C6 45:18 — **But you first wanna put a clamp right in here, see.**
- c) C6 00:24 — **You really don't want to go too much into the crease.**

We're a team 13.5%

Surgeon uses *we/us* instead of *you*.

- a) C3 08:31 — So, **we'll** put in a 10mm port.
- b) C6 18:49 — So now **we** are going to go around.
- c) C3 06:26 — **Let's** take a look in the upper abdomen.
- d) C7 35:00 — **Let's** go straight back.

What I do 8.2%

Surgeon says what they would or would not do in that context.

- a) C6 04:41 — So, I **would** try to go around here.
- b) C3 12:13 — No, I **like to** do it over an endocatch bag.
- c) C6 05:40 — So, the **best thing to do is just** get in and see...

Three classes of implicature

- ▶ **Gricean relevance**

Partial instruction that lets listener infer full instruction

- ▶ **Social connection**

Instruction is stated but embedded under 1st/2nd person pronouns

- ▶ **Modal embedding**

Instructions are softened under a modal expression

Ability 8.5%

Surgeon uses ability or other possibility modal

- a) C1 03:47 — **Can** you pull this here at the top?
- b) C2 05:27 — You **can** cut a little more.
- c) C3 44:24 — We **can** extend this a little bit....

Sometimes these are actually warnings

- d) C1 14:30 — You **can** go tangential there. (= DON'T go tangential)

Obligation 7.6%

Surgeon uses necessity modal

- a) C6 33:05 — You **have to** clamp both sides.
- b) C2 08:09 — You've **got to** be careful, poking around in there.
- c) C3 47:25 — You'll **need** a towel here....

Others

Attitude Softening 3.4%

Surgeon embeds command under *think* or *want*

- a) C3 22:00 — The incision, I'd like you to make a little bigger.
- b) C8 48:17 — **I want you to get the anterior.**

Others

Attitude Softening 3.4%

Surgeon embeds command under *think* or *want*

- a) C3 22:00 — The incision, I'd like you to make a little bigger.
- b) C8 48:17 — I want you to get the anterior.

Conditional 2.2%

Surgeon uses *if*-clause, often as a warning

- c) C6 05:17 — If you can clean this up, we're good.
- d) C1 03:33 — If you stand there, they won't see anything we're doing.
[don't stand there]

Multiple-in-one cases

Sometimes strategies stack together.

a) C6 30:21 — I think that's what you're going to have to do

Attitude Softening + Modal Embedding

b) C2 03:27 — Instead of switching to your scissors, we can just pass the fallopian tube

We're a team+ Ability

c) C8 32:35 — We still should just do this as a supercervical

We're a team + Obligation

Variety of indirect strategies

Gricean relevance	46.4%	Vague assent, Observation, Steps, Ellipsis
Social connection	31.1%	What you want, We're a team, What I do
Modal embedding	21.6%	Ability, Obligation, Conditional, Attitude softening
(other)	0.9%	Why don't, Analogy

Pedagogical value

Gricean relevance	Gets residents thinking
Social connection	Builds rapport
Modal embedding	Teaches general principles

Beyond surgical teaching

- ▶ Surgeons do speak pretty normally. Liu *et al.* 2021
- ▶ Should we see these rates and types elsewhere?
- ▶ Are other teachers this implicit?
- ▶ Is this too much implicit instruction?
- ▶ How can we better ensure understanding?

The end



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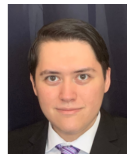
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References

Grice, H. P. 1975. Logic and conversation. In P. Cole, & J. Morgan (Eds.), *Syntax and semantics* (pp. 41-58). New York, NY: Academic Press.

Liu, C., McKenzie, A., and Sutkin, G. 2021. Semantically Ambiguous Language in the Teaching Operating Room. *Journal of Surgical Education*. 78:6, 1938–1947.

Mahadev, A., Bachar, A., Karmarkar, T., McKenzie, A., and Sutkin, G. 2023. Implicit communication and miscommunication in surgical instruction. *Global Surg Educ* 2, 88.

Slides:

